

Classic Theories of Economic Growth and Development

Chapter 3

Econ 322 – The Global Economy: Trade and Development

University of Tennessee, Knoxville

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

Outline

- ▶ **Classic Theories of Economic Development: Four Approaches**
- ▶ Development as Growth and the Linear-Stages Theories
- ▶ Structural-Change Models
- ▶ The International-Dependence Revolution
- ▶ The Neoclassical Counter-Revolution: Market Fundamentalism
- ▶ Classic Theories of Development: Reconciling the Differences

Classic Theories of Economic Development: Four Approaches

- ▶ Linear stages of growth model
- ▶ Theories and patterns of structural change
- ▶ International-dependence “revolution”
- ▶ Neoclassical, free market “counterrevolution”

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- ▶ Classic Theories of Economic Development: Four Approaches
- ▶ **Development as Growth and the Linear-Stages Theories**
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- ▶ Classic Theories of Development: Reconciling the Differences

Development as Growth and Linear-Stages Theories (1/2)

- ▶ **Post-WWII** Context
- ▶ Surge of interest in low-income countries after WWII and decolonization
- ▶ Industrialized nations lacked frameworks to analyze agrarian economies
- ▶ Inspiration came from:
 - ▶ The Marshall Plan (U.S. aid helped rebuild Europe rapidly)
 - ▶ Historical industrialization of today's developed countries



Marshall Plan poster (1950): "Whatever the weather, we only reach welfare together".

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Development as Growth and Linear-Stages Theories (2/2)

- ▶ **Core Ideas of Linear-Stages Theory**
- ▶ Development = Replicating the growth path of the West
- ▶ Emphasized massive capital injection (aid, investment, savings)
- ▶ Promoted belief in a universal sequence of growth stages
- ▶ **“Capital Fundamentalism”**
- ▶ Growth is driven by accelerated capital accumulation
- ▶ Assumes capital is the key constraint in poor countries
- ▶ Widely accepted by early development scholars and policymakers

Rostow's Stages of Growth (1/2)

Walt Rostow's Theory of Economic History

- ▶ “It is possible to identify all societies... within one of five categories.” — Rostow, *The Stages of Economic Growth* (1960)
- ▶ All economies develop in a linear, historical sequence: every country passes through the same five stages, from traditional society to modern industrial power

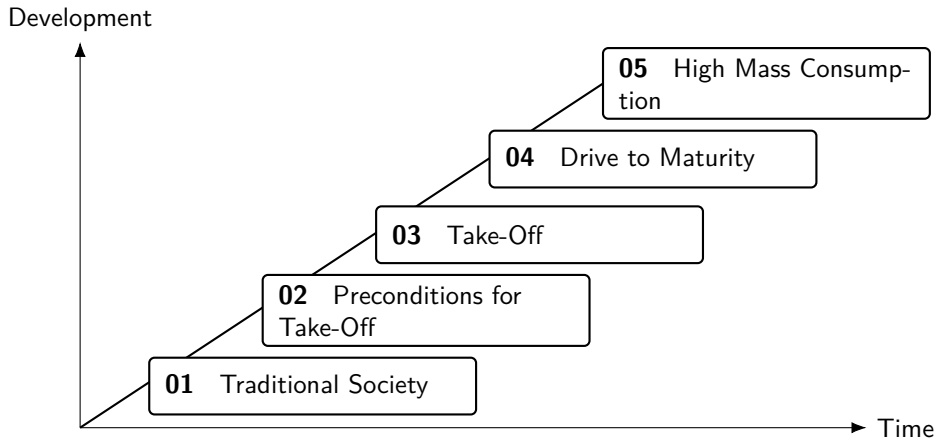
Takeaway: Rostow's model is **linear**, **universal**, and **investment-driven**; the key to take-off is capital accumulation.



Walt W. Rostow (1916–2003), economist and U.S. national security adviser, 1968.

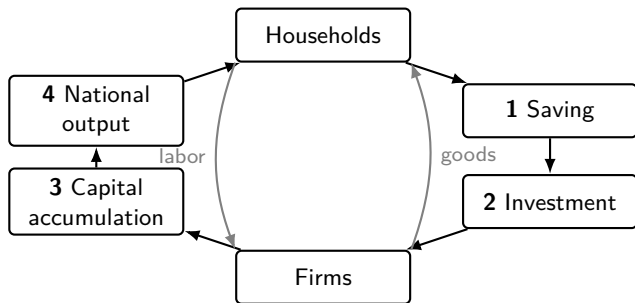
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Rostow's Stages of Growth (2/2)



Harrod-Domar Growth Model

- ▶ Sometimes called the “AK model”
- ▶ **Core Assumptions**
- ▶ Economies must **save and invest** to grow
- ▶ Investment = Net addition to **capital stock**
- ▶ Capital stock determines national **output (GDP)** through a **capital-output ratio (c)**



Simplified Harrod-Domar Growth Model (1/3)

► Harrod-Domar Growth Model Derivation

We start with the assumption of a **linear relationship** between output and capital:

$$Y = \frac{1}{c} \cdot K$$

Where:

- Y = Output (GDP)
- K = Capital stock
- c = Capital-output ratio (ICOR: Incremental Capital-Output Ratio)

Simplified Harrod-Domar Growth Model (2/3)

► Harrod-Domar Growth Model Derivation

Taking the **change in output**:

$$\Delta Y = \frac{1}{c} \cdot \Delta K$$

Since investment equals change in capital:

$$\Delta Y = \frac{S^{\text{Net}}}{c}$$

Now divide both sides by Y :

$$\frac{\Delta Y}{Y} = \frac{S^{\text{Net}}}{Y \cdot c}$$

Recognizing that $\frac{S^{\text{Net}}}{Y} = s^{\text{Net}}$ (net savings rate), we have:

$$\text{Growth} = \frac{s^{\text{Net}}}{c}$$

Simplified Harrod-Domar Growth Model (3/3)

Implications of $\frac{\Delta Y}{Y} = \frac{s}{c}$

- ▶ Higher saving rate $s \Rightarrow$ more investment $\Rightarrow$ faster growth
- ▶ Lower capital-output ratio $c \Rightarrow$ more efficient use of capital $\Rightarrow$ faster growth

The big picture: save more, invest better, and you will grow faster. This idea was very influential in early development planning: governments and international organizations believed growth could be engineered by raising savings and investment rates.

The Harrod-Domar Model – Simplified Version, Alternate Derivation (1/2)

1. Savings Function:

$$S = sY \quad (3.1)$$

2. Investment Definition:

$$I = \Delta K \quad (3.2)$$

3. Capital-Output Relationship:

$$\Delta K = c \cdot \Delta Y \quad (3.3)$$

4. Savings = Investment:

$$S = I \quad (3.4)$$

The Harrod-Domar Model – Simplified Version, Alternate Derivation (2/2)

Combine (3.1)–(3.4):

$$sY = c \cdot \Delta Y = \Delta K = I \quad (3.5)$$

Rearranging:

$$sY = c \cdot \Delta Y \quad (3.6)$$

Divide both sides by Y :

$$\frac{\Delta Y}{Y} = \frac{s}{c} \quad (3.7)$$

Harrod-Domar with Capital Depreciation (1/3)

Step 1: Modified Growth Equation

$$\frac{\Delta Y}{Y} = \frac{s^G}{c} - \delta$$

Where:

- ▶ s^G : Gross savings rate
- ▶ c : Capital-output ratio (ICOR)
- ▶ δ : Capital depreciation rate
- ▶ $\frac{\Delta Y}{Y}$: Growth rate of GDP

Harrod-Domar with Capital Depreciation (2/3): Example

If:

- ▶ Savings rate $s^G = 20\%$
- ▶ ICOR $c = 4$
- ▶ Depreciation $\delta = 5\%$

Then growth is:

$$\frac{\Delta Y}{Y} = \frac{0.20}{4} - 0.05 = 0.05 - 0.05 = 0$$

All savings go to maintaining the capital stock — there is no net growth. Investment must be high enough not just to expand the capital stock but also to replace worn-out assets.

Harrod-Domar with Capital Depreciation (3/3): Derivation

$$c = \frac{\Delta K}{\Delta Y} \Rightarrow \Delta Y = \frac{1}{c} \Delta K$$

Capital accumulation with depreciation:

$$\Delta K = I^G - \delta K$$

Substitute into the output equation:

$$\Delta Y = \frac{1}{c} (I^G - \delta K) = \frac{1}{c} I^G - \frac{1}{c} \delta K$$

Assume $I^G = S^G$ and $S^G/Y = s^G$; divide by Y :

$$\frac{\Delta Y}{Y} = \frac{s^G}{c} - \delta$$

Illustrative Exercise: Growth Potential

Case 1: Lower Savings Scenario

Suppose:

$$s^G = 0.125, \quad \delta = 0.04, \quad c = 2.5$$

Then:

$$g^{TOT} = \frac{0.125}{2.5} - 0.04 = 0.01$$

Case 2: Higher Savings Scenario

But if:

$$s^G = 0.175, \quad \delta = 0.03, \quad c = 2.5$$

Then:

$$g^{TOT} = \frac{0.175}{2.5} - 0.03 = 0.04$$

Small improvements in the savings rate and the depreciation rate can lead to significantly higher growth: here growth quadruples from 1% to 4%.

Illustrative Exercise: Growth Targets

Assumptions: gross savings rate **20%**; depreciation rate **3%**; ICOR (incremental capital-output ratio) **2.5**.

1. Using the Harrod-Domar growth model, find the implied GDP growth rate:

$$\text{Growth} = \frac{s}{\text{ICOR}} - \delta = \frac{0.2}{2.5} - 0.03 = 0.08 - 0.03 = \boxed{0.05 \text{ or } 5\%}$$

2. How much would the savings rate need to increase to reach a 9% growth target?

Let the required savings rate be s . Solve for s when target growth is 9%:

$$0.09 = \frac{s}{2.5} - 0.03 \Rightarrow \frac{s}{2.5} = 0.12 \Rightarrow s = 0.3 \text{ or } 30\%$$

Solow Growth Model

- ▶ Why do we study this model?
- ▶ The general question in growth models:
 - ▶ (i) what are the determinants of long run economic growth?
 - ▶ (ii) how can we explain the vast difference in both output and growth rates across countries/time?
- ▶ Solow's specific question: what do simple neoclassical assumptions imply about growth?
- ▶ We will study the Solow growth model in detail; first we need its basic structure and underlying assumptions



Robert M. Solow (1924–2023), MIT; Nobel Prize 1987 for his theory of growth.

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Solow Growth Model

Key assumptions

- ▶ Constant returns to scale
- ▶ Perfect competition
- ▶ Perfect information
- ▶ No externalities
- ▶ No government (in our initial setting, we will relax later)
- ▶ Closed economy
- ▶ The economy saves at a constant rate (determined exogenously)

The basic structure

- ▶ We consider a composite unit – a household/producer. It produces

$$Y(t) = F[K(t), L(t), t] \quad (\#1)$$

Solow Growth Model

- ▶ Two inputs: physical capital $K(t)$ and labor $L(t)$. The production function also depends on time t – to reflect the effects of technological progress
- ▶ One sector production technology
 - ▶ output is a homogeneous good
 - ▶ The good could be consumed $c(t)$ or invested $I(t)$ to create new physical capital $K(t)$
- ▶ Let $s(\cdot)$ be the fraction of output that is saved ($\Rightarrow$ savings rate).
- ▶ $1 - s(\cdot)$ is the fraction of output that is consumed in the current period
- ▶ Solow (1956) and Swan (1956) assume s to be constant (given exogenously)
- ▶ Assume capital depreciates at a constant rate $\delta > 0$ (capital wears out)

Solow Growth Model

- ▶ Hence, net increases in the stock of physical capital at a point of time:

$$\dot{K} = I - \delta K = sF(K, L, t) - \delta K \quad (\#2)$$

- ▶ Here, $\dot{K} = \frac{dK}{dt}$ (is the differentiation of K with respect to time)
- ▶ For simplification, for the time being, assume no technological progress. So $F(\cdot)$ is independent of time.
- ▶ Assume that the population grows at a constant exogenous rate $\frac{\dot{L}}{L} \geq 0$
- ▶ Also, assume that everybody works at a given intensity

Solow Growth Model

- ▶ Normalize population at time 0 to 1
- ▶ Then:

$$L(t) = L_0 e^{nt} = 1 \cdot e^{nt} = e^{nt} \quad (\#3)$$

The Simple Model

- ▶ The production function is:

$$Y = F(K, L) \quad (\#4)$$

- ▶ The production function is neoclassical if the following three properties are satisfied:
 - First, $\forall K > 0$ and $L > 0$, $F(\cdot)$ exhibits positive and diminishing marginal product with respect to each input: $\frac{\partial F}{\partial K} > 0$, $\frac{\partial^2 F}{\partial K^2} < 0$, $\frac{\partial F}{\partial L} > 0$, $\frac{\partial^2 F}{\partial L^2} < 0$

Solow Growth Model

- First, $\forall K > 0$ and $L > 0$, $F(\cdot)$ exhibits positive and diminishing marginal product with respect to each input:

$$\frac{\partial F}{\partial K} > 0, \quad \frac{\partial^2 F}{\partial K^2} < 0, \quad \frac{\partial F}{\partial L} > 0, \quad \frac{\partial^2 F}{\partial L^2} < 0 \quad (5.1)$$

- Second, $F(\cdot)$ exhibits constant returns to scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L) \quad \forall \lambda > 0 \quad (5.2)$$

- Third, the marginal product of capital (or labor) approaches infinity as capital (or labor) goes to 0 and approaches 0 as capital (or labor) goes to infinity:

$$\lim_{K \rightarrow 0} F_K = \lim_{L \rightarrow 0} F_L = \infty \quad \text{and} \quad \lim_{K \rightarrow \infty} F_K = \lim_{L \rightarrow \infty} F_L = 0 \quad (5.3)$$

Solow Growth Model

- Constant Returns to Scale (CRS) implies:

$$Y = F(K, L) = LF\left(\frac{K}{L}, 1\right) = Lf(k)$$

- Where $k \equiv \frac{K}{L}$ is the capital-labor ratio, $y \equiv \frac{Y}{L}$ is the per capita output and $F(k, 1) \equiv f(k)$

Intensive form:

$$y = f(k) \tag{#6}$$

- Recall, $Y = Lf(k) = Lf\left(\frac{K}{L}, 1\right)$ and differentiate with respect to K , for fixed L :

$$\frac{\partial Y}{\partial K} = Lf'(k)\frac{1}{L} = f'(k) \tag{7.1}$$

Solow Growth Model

- Differentiate with respect to L , for given K :

$$\frac{\partial Y}{\partial L} = [f(k) - kf'(k)] \quad (7.2)$$

- Note that equations (5.1)–(5.3) imply that each input is essential for production, that

$$\begin{aligned} F(0, L) &= F(K, 0) = f(0) = 0 \\ F(\infty, L) &= F(K, \infty) = f(\infty) = \infty \end{aligned}$$

- Example: Cobb Douglas production function:

$$Y = AK^\alpha L^{1-\alpha}, \quad A > 0, \quad 0 < \alpha < 1 \quad (\#8)$$

Solow Growth Model

- ▶ The intensive form:

$$y = Ak^{\alpha} \quad (\#9)$$

- ▶ Note that $f'(k) = A\alpha k^{\alpha-1} > 0$; $f''(k) = -A\alpha(1-\alpha)k^{\alpha-2} < 0$; $\lim_{k \rightarrow \infty} f'(k) = 0$ and $\lim_{k \rightarrow 0} f'(k) = \infty$
- ▶ The Cobb-Douglas form satisfies the properties of a neoclassical production function

Solow Growth Model

The Fundamental Dynamics Equation for the Capital Stock

- ▶ We will analyze the dynamic behavior of the economy described by the neoclassical production function
- ▶ Recall equation (#2):

$$\dot{K} = sF(K, L) - \delta K \implies \frac{\dot{K}}{L} = sf(k) - \delta k$$

- ▶ Note: the RHS variables are in per capita terms, while LHS variables are not
- ▶ We can write:

$$\dot{k} = \frac{d\left(\frac{K}{L}\right)}{dt} = \frac{\dot{K}L - K\dot{L}}{L^2} = \frac{\dot{K}}{L} - nk \implies \frac{\dot{K}}{L} = \dot{k} + nk$$

- ▶ Substitute this into the above equation to get:

Solow Growth Model

► Substitute this into the above equation to get: $\dot{k} + nk = sf(k) - \delta k$

► Rearranging, we have:

$$\dot{k} = sf(k) - (n + \delta)k \quad (\#10)$$

► Equation (#10) is the fundamental differential equation of the Solow-Swan model. This non-linear equation depends only on k

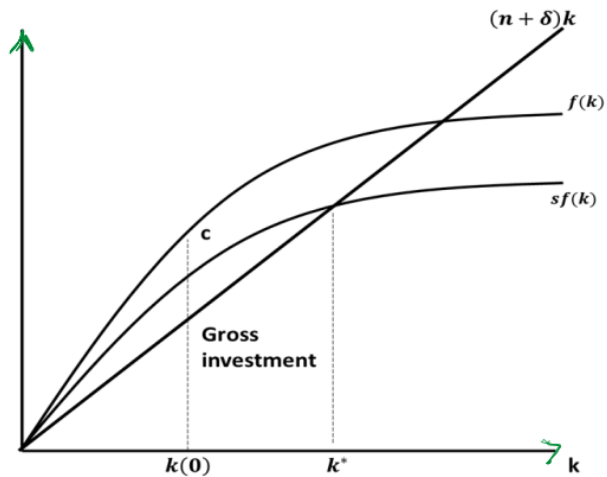
► $n + \delta$ is the effective depreciation rate for the capital-labor ratio $k = \frac{K}{L}$

► $f(k)$ production function and

► $sf(k)$ savings function

► $f(0) = 0$ has a positive slope $f'(k) > 0$ gets flatter as k rises $f''(k) < 0$

Solow Growth Model



Solow Growth Model

- ▶ Consider an economy with initial $k(0) > 0$ (the diagram on the last page)
- ▶ Note that at $k(0) \rightarrow sf(k) > (n + \delta)k$
- ▶ $c = (1 - s)f(k)$
- ▶ At $k(0) \rightarrow \dot{k} = sf(k) - (n + \delta)k > 0$
- ▶ Investment is positive $\Rightarrow k$ goes up
- ▶ Now, we will discuss about the equilibrium in this economy
- ▶ We use the term “steady state” to indicate long run equilibrium

Solow Growth Model

Steady State (SS)

- ▶ Steady state is a situation in which the variables grow at a constant rate
- ▶ In this model, the SS implies that $\frac{\dot{k}}{k} = 0$
- ▶ Note that k in the diagram at which $sf(k) = (n + \delta)k \rightarrow k^*$
- ▶ This implies that:

$$sf(k^*) = (n + \delta)k^* \quad (\#11)$$

Solow Growth Model

- ▶ Prove that k must be constant in the steady state $\Rightarrow \dot{k} = 0$ at the SS
- ▶ Recall (#10):

$$\dot{k} = sf(k) - (n + \delta)k \rightarrow \frac{\dot{k}}{k} = \frac{sf(k)}{k} - (n + \delta)$$

- ▶ At the steady state, $\frac{\dot{k}}{k}$ is constant (by definition)
- ▶ Also, note that s, δ, n are constant in the model
- ▶ Therefore, $f(k)/k$ must be constant in the steady state
- ▶ By taking time-derivative:

Solow Growth Model

- By taking time-derivative we get:

$$\frac{d}{dt} \left[\frac{f(k)}{k} \right] = 0 \rightarrow \frac{f'(k) \dot{k} k - \dot{k} f(k)}{k^2} = -\frac{1}{k} [f(k) - k f'(k)] \frac{\dot{k}}{k} = - \left(\frac{\partial Y}{\partial L} \right) \frac{1}{k} \frac{\dot{k}}{k} = 0$$

- We know, $\frac{\partial Y}{\partial L} > 0$ and, as long as $\frac{K}{L} = k$ is finite, the steady state implies that $\frac{\dot{k}}{k} = 0$ (proved).
- Note that we focus on the $k > 0$ zone. Since k is constant in the steady state (k^*):
- $y^* = f(k^*)$ is constant and $c^* = (1 - s)f(k^*)$ is constant

Solow Growth Model

- ▶ Note that in the neoclassical model, the per capita quantities k, y, c do not grow in the steady-state
- ▶ The levels of the variables K, Y, C grow in the steady state at the rate of the population growth, n .

Changes:

- ▶ 1) in the level of technology: shift the production function $f(\cdot)$. If positive, it shifts upwards and k increases.
- ▶ 2) increases in s : shifts $sf(k)$ upward and k increases.
- ▶ 3) increases in δ or n : moves the $(n + \delta)k$ line upward and k decreases.

Solow Growth Model

- ▶ Note 1: Technology, s , n , δ do not affect the steady state growth rate of per capita output, capital and consumption (all of which are equal to zero)
- ▶ Note 2: The model as specified will not provide explanations of the determinants of long-run per capita growth

The Golden Rule of Capital Accumulation and Dynamic Inefficiency

- ▶ We know for given values of n and δ , there is a unique steady-state value k^* for each level of savings rate s
- ▶ Note $k^*(s)$ with $\frac{dk^*(s)}{ds} > 0$

Solow Growth Model

- ▶ The steady state level of per capita consumption:

$$c^* = (1 - s)f[k^*(s)]$$

- ▶ Recall, equation (#11): $sf(k^*) = (n + \delta)k^*$ (at SS)

- ▶ Hence, we can write:

$$c^* = (1 - s)f[k^*(s)] = f[k^*(s)] - (n + \delta)k^*(s) \quad (\#12)$$

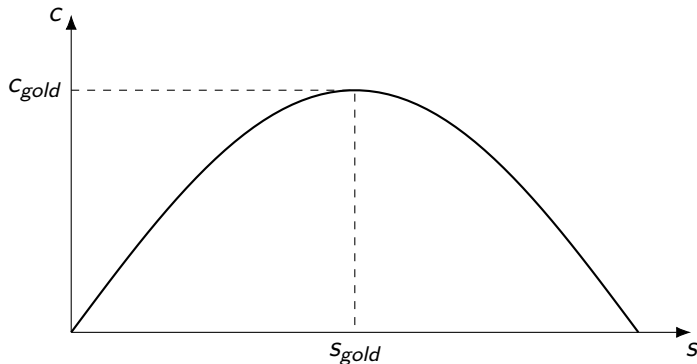
- ▶ The figure shows the relationship between c^* and s that is implied by:

$$c^* = (1 - s)f[k^*(s)] = f[k^*(s)] - (n + \delta)k^*(s)$$

- ▶ c^* is increasing in savings for low levels of s and decreasing in s for high levels of s

Solow Growth Model

c^* is increasing in savings for low levels of s and decreasing in s for high levels of s



Solow Growth Model

- ▶ There is c^* reaching at its maximum level when first derivative is zero:

$$\frac{dc^*(s)}{ds} = 0 \Rightarrow [f'(k^*) - (n + \delta)] \frac{dk^*}{ds} = 0$$

- ▶ Note $\frac{dk^*}{ds} > 0 \Rightarrow f'(k^*) - (n + \delta) = 0$ OR, $f'(k^*) = n + \delta$
- ▶ We will denote this $k^* \rightarrow k_{gold}$. Then:

$$f'(k_{gold}) = n + \delta \quad (\#13)$$

- ▶ The corresponding savings rate is s_{gold} and per-capita consumption is c_{gold}

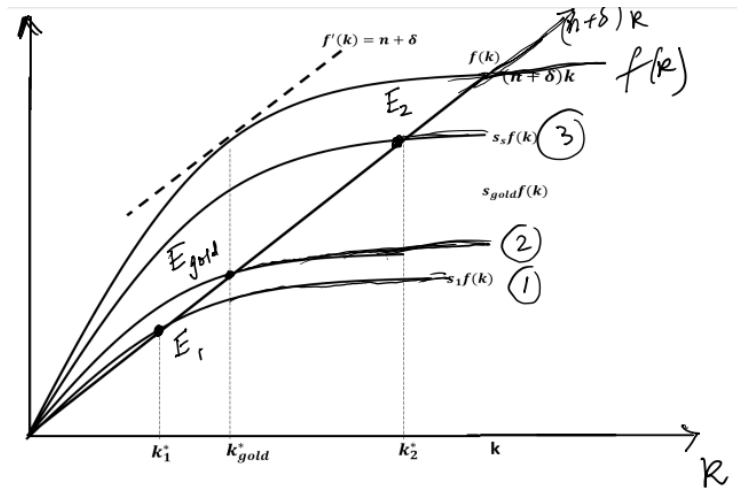
Solow Growth Model

- ▶ Following equation (#12):

$$c_{gold} = f(k_{gold}) - (n + \delta)k_{gold}$$

- ▶ Equation (#13) is the golden rule of capital accumulation.
- ▶ The following figure illustrates the workings of golden rule
- ▶ Consider three possible saving rates, $s_1 < s_{gold} < s_2$
- ▶ Per capita consumption c in each case equals the vertical distance between the production function $f(k)$ and the corresponding $sf(k)$ curve
- ▶ For each s , the steady-state value k^* corresponds to the intersection between the $sf(k)$ curve and the $(n + \delta)k$ line

Solow Growth Model



Solow Growth Model

- ▶ Since $s_1 < s_{gold} < s_2 \rightarrow k_1^* < k_{gold} < k_2^*$
- ▶ Unless we have a specific objective function, we cannot say which saving rate is the best one for the economy
- ▶ However, we can argue that a saving rate that exceeds s_{gold} is inefficient because higher quantities of per capita consumption could be obtained at all points in time by reducing the savings rate
- ▶ c^* is maximized when $k^* = k_{gold}$ because the slope of the production function is parallel to $(n + \delta)k$ line: $f'(k) = n + \delta$
- ▶ If the savings rate is below the golden rule $s_1 < s_{gold}$, then an increase in s will increase the steady-state consumption per person but lower consumption per person along the transition.

Necessary vs. Sufficient Conditions: Criticisms of the Stages Model

Key argument: saving and investment are **necessary but not sufficient** for economic growth.

Why the stages model sometimes fails

- ▶ The Marshall Plan succeeded in Europe due to:
 - ▶ Well-functioning markets
 - ▶ Efficient transport systems
 - ▶ Educated workforce
 - ▶ Motivated population
 - ▶ Competent government bureaucracy
- ▶ These structural, institutional, and attitudinal conditions were often missing in underdeveloped countries

Unrealistic assumptions in Rostow and Harrod-Domar

- ▶ Assumed that underdeveloped countries had:
 - ▶ Skilled labor
 - ▶ Managerial competence
 - ▶ Capacity to implement development projects
- ▶ In reality, these were often absent

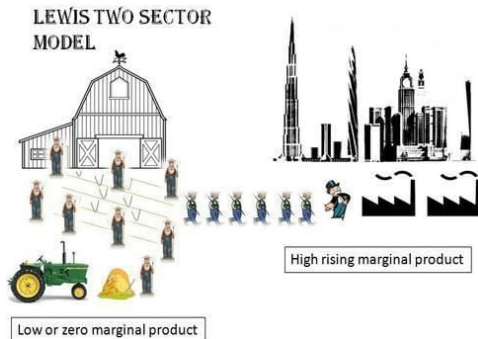
Outline

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- ▶ Development as Growth and the Linear-Stages Theories
- ▶ **Structural-Change Models**
- ▶ The International-Dependence Revolution
- ▶ The Neoclassical Counter-Revolution: Market Fundamentalism
- ▶ Classic Theories of Development: Reconciling the Differences

Structural-Change Models

Definition: structural-change theory examines how **underdeveloped economies** evolve from

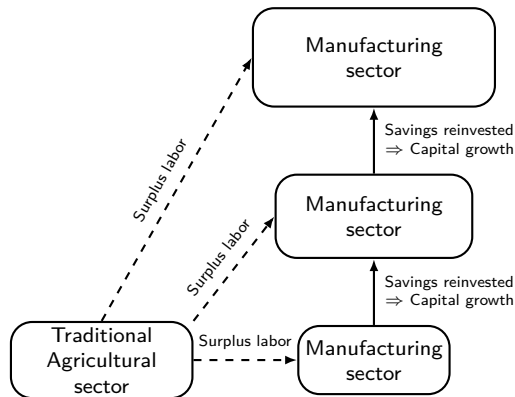
- ▶ **traditional subsistence agriculture** →
 - ▶ **modern, urbanized economies** with diverse **manufacturing** and **service** sectors
-
- ▶ Development as a reallocation of labor and resources: low- to high-productivity sectors, rural areas to cities, informal to formal wage employment
 - ▶ Structural change is essential, but it must be **productive**, not just geographic



Surplus labor with low marginal product moves from the farm to the modern sector.

The Lewis two-sector model (1/4)

- ▶ Developed by **W. Arthur Lewis** (1950s); extended by **Fei and Ranis**
- ▶ Focus: Structural transformation from a **subsistence rural sector** to a **modern urban industrial sector**
- ▶ One of the most influential models for **surplus-labor economies**



The Lewis two-sector model (2/4)

- ▶ **Structure of the Economy in the Lewis Model**
- ▶ **Two-sector growth model:** agriculture and industry
- ▶ **Industry:** neoclassical/Solow sector $\Rightarrow$ growth through investment in capital
 - *But:* profit (and thus investment) in industry depends on agriculture through use of food and labor
- ▶ **Aim:** show how agricultural development determines the speed of industrialization
- ▶ **Intuition:** low-productive agricultural labor can be more productive in industry provided food production remains adequate

The Lewis two-sector model (3/4)

- ▶ **Key Assumptions**
- ▶ **Surplus labor** exists in the rural sector ($MPL = 0$)
- ▶ **Capitalists reinvest all profits** in modern sector
- ▶ **Urban wage** (W_M) is fixed above rural wage (W_A)
→ **Perfectly elastic labor supply** to modern sector at W_M
- ▶ Modern sector **hires labor until** $MPL = W_M$

The Lewis two-sector model (4/4)

Growth Dynamics

Profits → Reinvestment → Capital Growth → Labor Demand

- ▶ Fixed wages plus rising output per worker create profits for the owners of capital
- ▶ Profits are fully reinvested in the modern sector: new factories, machines, infrastructure
- ▶ A larger capital stock raises labor demand; surplus labor keeps arriving from agriculture at the fixed urban wage

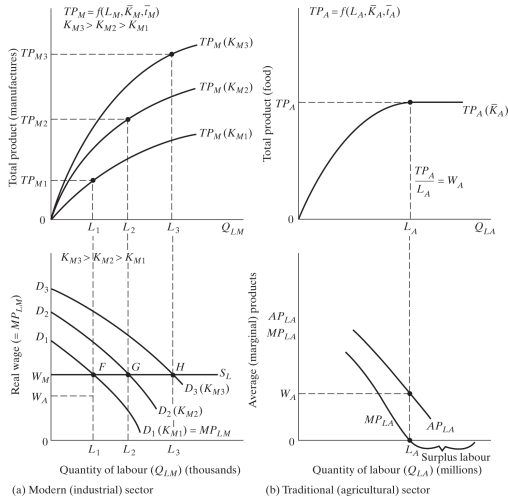
A virtuous circle of reinvestment and employment growth... until surplus labor is exhausted and real wages start to rise (the **Lewis turning point**).



Bust of W. Arthur Lewis (1915–1991), Saint Lucia; Nobel 1979.

Wikimedia, Ulises Icardi, CC BY-SA 4.0

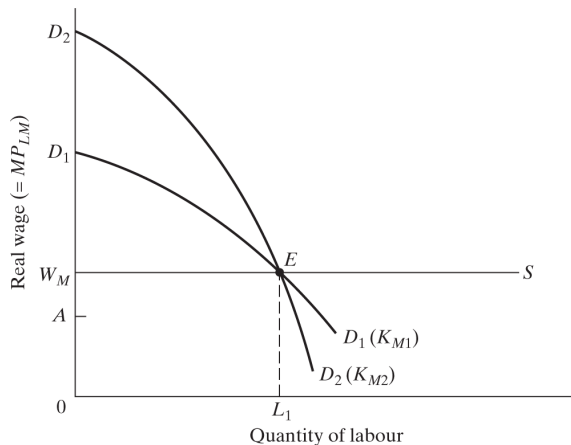
Figure 3.1 The Lewis Model



Criticisms of the Lewis Model

- ▶ The rate of labor transfer and employment creation may not be proportional to the rate of modern-sector capital accumulation
- ▶ Questionable assumption of surplus labor in rural areas and full employment in urban areas
- ▶ Questionable assumption of diminishing returns in the modern industrial sector, at least in some cases
- ▶ Institutional factors have a small (if any) role in this approach

Figure 3.2 The Lewis Model Modified by Laborsaving Capital Accumulation: Employment Implications



Empirical Tests and Practical Applications of the Lewis Model

► Empirical limitations

- Evidence does not consistently support the existence of surplus labor with zero marginal productivity.
- Difficult to quantify the amount of surplus labor and its impact on output.

► Case: India (Schultz, 1918–19)

- Agricultural output declined during labor shortages caused by the influenza epidemic.
- Suggests that labor in agriculture may have positive marginal productivity.

► Case: Egypt (Mabro, 1967)

- Model failed due to underestimated population growth.
- Industrial policy favored capital-intensive methods, limiting labor absorption.
- Unemployment remained high despite structural change.

► Case: Taiwan

- Strong economic growth did not reduce unemployment significantly.
- Capital-intensive industrialization again limited the transfer of labor from agriculture.

Empirical Patterns of Development - Examples

- ▶ Some processes are very typical but not universal; so, these are often referred to as “stylized facts”:
 - ▶ Switch from **agriculture to industry** (and services)
 - ▶ **Rural-urban migration** and urbanization
 - ▶ Steady accumulation of **physical and human capital**
 - ▶ **Population growth** first increases and then decreases with a **decline in family size**

Stylized facts are not universal laws but regularities that help us understand how development typically unfolds: rising productivity, changing demand, urbanization, and the demographic transition.

Conclusions and Implications of Structural-Change Models

- ▶ Development follows common patterns, such as:
 - ▶ Shift from agriculture to industry and services
 - ▶ Urbanization and capital accumulation
- ▶ Patterns vary across countries due to:
 - ▶ Resource endowment, size, policy, access to capital/technology, trade environment
- ▶ Caution: Patterns $\neq$ Causes
 - ▶ Imitating outcomes (e.g., urbanization, universities) without foundational conditions (e.g., basic literacy, agricultural support) can lead to policy mistakes
- ▶ Optimistic view:
 - ▶ The right policy mix can lead to self-sustaining growth
- ▶ Sets the stage for the next theory:
 - ▶ International-dependence models challenge this optimism

Outline

- ▶ Classic Theories of Economic Development: Four Approaches
- ▶ Development as Growth and the Linear-Stages Theories
- ▶ Structural-Change Models
- ▶ **The International-Dependence Revolution**
- ▶ The Neoclassical Counter-Revolution: Market Fundamentalism
- ▶ Classic Theories of Development: Reconciling the Differences

The International-Dependence “Revolution” (1/4)

- ▶ **Overview**
- ▶ Emerged in the 1970s, reflecting frustration with stages and structural-change models
- ▶ Gained support among developing-country scholars
- ▶ Resurged in the 21st century in anti-globalization movements



The International-Dependence “Revolution” (2/4)

- ▶ **Core Viewpoint**
- ▶ Developing countries are constrained by:
 - ▶ Institutional, political, and economic rigidities
 - ▶ Dependence on and dominance by richer nations
- ▶ The IDR models reject the emphasis on GNP growth rate as the principal index of development.
- ▶ IDR models view developing countries as beset by institutional, political, and economic rigidities — both domestic and international.
- ▶ IDR models argue that developing countries are in a dominant and dependent relationship with rich countries.

The International-Dependence “Revolution” (3/4)

- ▶ **Key Concepts**
- ▶ Dependence:
- ▶ Developing countries rely on developed nations for:
 - ▶ Economic policies
 - ▶ Technology, education, culture, and consumption models
- ▶ Dominance:
- ▶ Rich countries have **disproportionate power** in:
 - ▶ Global trade and pricing (e.g., for raw materials)
 - ▶ Political and financial systems

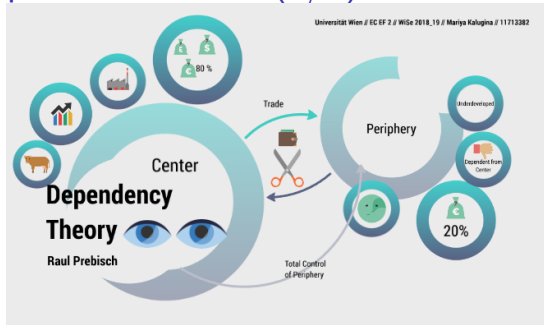
The International-Dependence “Revolution” (4/4)

Three Major Schools of Thought

1. **Neocolonial Dependence Model:** even after independence, developing countries remain economically dependent on rich nations and multinational corporations; domestic elites benefit from the status quo and block reform — “colonialism without the flag”
2. **False-Paradigm Model:** development fails because countries follow flawed models pushed by external advisors that ignore local realities and impose one-size-fits-all solutions
3. **Dualistic-Development Thesis:** a modern, urban, high-productivity sector coexists with a traditional, rural, low-productivity sector, and the gap tends to widen rather than close

All three challenge the idea that development failure is simply about poor choices by poor countries: systemic forces, internal and external, keep developing nations trapped.

The Neocolonial Dependence Model (1/5): Global Power



Center and periphery in dependency theory (Raúl Prebisch): trade ties the periphery to the center, which keeps control.

Even after formal colonialism ended, many countries remained economically and politically dependent on richer nations — a condition that continues to hold back development.

The Neocolonial Dependence Model (2/5): Background and Core Idea

Background

- ▶ Former colonies gained independence post-WWII
- ▶ Colonial powers left, but influence remained
- ▶ Economic control replaced direct political control: trade patterns still favored former colonial powers, export profits flowed outward, key decisions were made by foreign investors

Core Idea

- ▶ Poor countries are dependent on rich countries
- ▶ Domestic elites maintain the dependency
- ▶ Underdevelopment is a result of external and internal forces

“Dual exploitation”: both global and domestic elites share responsibility for underdevelopment — colonialism in disguise.

The Neocolonial Dependence Model (3/5): Features and Elites

Key Features

- ▶ Continued economic dependence: exports of primary products whose prices are unstable; value-added happens elsewhere
- ▶ Foreign aid with strings attached: IMF/World Bank loan conditions (budget cuts, trade opening, privatization)
- ▶ Cultural and ideological influence: Western education, media, and consumer goods set the standard

Role of Elites in the Dependency System

- ▶ Local elites benefit from global linkages
- ▶ Act as intermediaries between foreign interests and local economies
- ▶ Often resist structural change (land reform, unions, taxing foreign firms)

The Neocolonial Dependence Model (4/5): Case and Relevance Today

Case – Francophone Africa

(Senegal, Côte d'Ivoire, Mali...)

- ▶ Used the French currency system (the CFA franc)
- ▶ Relied on French military and technical advisors
- ▶ Exported mainly to France
- ▶ In return, France kept access to resources and political influence

Relevance Today

- ▶ Global trade and supply chains lock poor countries into low-value stages of production
- ▶ Corporate tax avoidance via tax havens and trade agreements
- ▶ Unequal access to technology and finance

Cocoa in West Africa → chocolate profits in Europe; textiles in Bangladesh → global brands capture most profits; coffee in Ethiopia → retailers earn far more than farmers.

The Neocolonial Dependence Model (5/5): Critiques and Comparison

Critiques

- ▶ Overemphasizes external blame, ignoring internal factors like corruption or mismanagement
- ▶ Understates the agency of developing countries, which make strategic choices
- ▶ Can be politically polarizing: a good-vs-bad narrative of international relations

Comparison with Other Dependency Models

Model	Focus
False Paradigm Model	Misapplied Western policies
Neocolonial Dependence Model	Structural global power
Dual-Sector Model (Lewis)	Internal labor dynamics

The False-Paradigm Model (1/9): Historical Context

Development failure through misguided policy advice: countries follow expert advice and implement the “right-looking” policies, yet remain poor, because the model itself is wrong for their situation.

- ▶ **Post-WWII independence** created urgency for rapid development
- ▶ **Western institutions** (IMF, World Bank) became dominant advisors
- ▶ Policies were based on **Western experience**, applied globally
- ▶ **Local realities** were often ignored: weak institutions, inequality, colonial legacies



Protest against IMF/World Bank policies.

The False-Paradigm Model (2/9): Core Idea

- ▶ Many countries **fail to develop** because they **adopt foreign-designed policies**
- ▶ These policies are based on **false assumptions** of market efficiency
- ▶ **Western-trained technocrats** often dominate national policymaking
- ▶ Result: **Mismatch** between policy design and actual constraints

Applying foreign policies without adapting them to local context leads to “**growth without development**”: GDP may rise, but poverty remains and inequality worsens.

The False-Paradigm Model (3/9): Assumptions and Critique

- ▶ **Neoclassical logic** assumes:
 - ▶ **Efficient markets**
 - ▶ **Perfect competition**
 - ▶ **GDP growth = success**
- ▶ **Neglects:**
 - ▶ **Inequality**
 - ▶ **Weak governance**
 - ▶ **Historical context**
- ▶ **Outcome: Growth without development**

The False-Paradigm Model (4/9): Common False Paradigms in Practice

Problem	Policy Advice	Result
Weak industry	Liberalize trade	Local firms collapse under foreign competition
Fiscal deficits	Austerity (cut spending)	Education & healthcare deteriorate
Low investment	Attract FDI with tax breaks	Profits repatriated, no local upgrading
Poverty & inequality	Focus on GDP growth	Social exclusion and unrest continue

When you import policies without adapting them, you risk making things worse.
Pick one row: how would you design a better solution, knowing the country's real constraints?

The False-Paradigm Model (5/9): Case 1 – Latin America

- ▶ 1980s: **Debt crisis** + **IMF intervention**
- ▶ Policies: **Structural Adjustment Programs**
 - ▶ Liberalization
 - ▶ Privatization
 - ▶ Austerity
- ▶ Results:
 - ▶ **“Lost Decade”**
 - ▶ Worsening **poverty** and **unemployment**
 - ▶ Public services declined

The False-Paradigm Model (6/9): Case 2 – Zambia

- ▶ Followed IMF advice:
 - ▶ **Privatized copper sector**
 - ▶ **Cut agricultural subsidies**
- ▶ Then:
 - ▶ **Copper prices crashed**
 - ▶ State revenue fell
 - ▶ **Food insecurity** increased
- ▶ Outcome: **Dependence and economic vulnerability**

The False-Paradigm Model (7/9): Academic Foundations

- ▶ **Paul Baran:** Global capitalism exploits poor nations; underdevelopment is not a natural state
- ▶ **André Gunder Frank:** Underdevelopment is **systemic**, not accidental; rich countries grow at the expense of poor countries
- ▶ **Gunnar Myrdal:** Institutions must change for development to occur — not just growth or aid
- ▶ **Dudley Seers:** Development = reduced **poverty, inequality, unemployment**; if these are not falling, can we call it development?

These thinkers shaped a different way of thinking about development — one that puts people and justice, not GDP, at the center.

The False-Paradigm Model (8/9): Strengths and Limitations

Strengths

- ▶ Emphasizes **institutional and historical context**
- ▶ Critiques **knowledge colonialism**
- ▶ Supports **endogenous, bottom-up development**

Limitations

- ▶ May **over-blame foreign influence**
- ▶ Offers few **concrete alternatives**
- ▶ Understates **internal mismanagement**

The False-Paradigm Model (9/9): Conclusion

- ▶ **False Paradigm Model** warns against **one-size-fits-all** strategies
- ▶ Effective development must be:
 - ▶ **Context-sensitive**
 - ▶ **Institutionally grounded**
 - ▶ **Locally informed**
- ▶ Calls for **intellectual independence** and **inclusive development paths**

Just because a policy worked in Germany or the U.S. does not mean it will work in Ghana or Nepal. Ask: who is giving the advice, what assumptions are being made, and does the policy fit the realities of the country?

The Dualistic-Development Thesis (1/5): Two Worlds, One Country?



Slums next to skyscrapers: a modern, globalized sector and a traditional, poor sector side by side.

In many poor countries two economies coexist — one modern and dynamic, the other traditional and stagnant — and the gap between them tends to grow rather than close, even when the economy grows.

The Dualistic-Development Thesis (2/5): What Is Dualism?

Dualism: two very different sectors exist in the same country, often right next to each other — a **modern** sector (urban, capital-intensive, globally connected) and a **traditional** sector (rural, labor-intensive, subsistence or informal work) — coexisting without integration, with inequalities in income, productivity, and opportunity.

Key Elements

1. Superior vs. inferior conditions coexist
2. No automatic trickle-down: growth in one sector does not necessarily help the other
3. Gaps may widen over time: the modern sector attracts the investment, talent, and attention
4. Coexistence is chronic, not transitional

The Dualistic-Development Thesis (3/5): Examples and Persistence

Examples of Dualism

- ▶ Urban-rural divides
- ▶ Formal vs. informal labor markets
- ▶ Modern export zones vs. subsistence farming
- ▶ Mumbai, Nairobi, Rio, Manila, Lagos: luxury districts next to informal settlements

Why Dualism Persists

- ▶ Skills gap and unequal education
- ▶ Infrastructure and location advantages concentrated in cities
- ▶ Political capture and urban bias

The existence of the modern sector does not by itself help the traditional sector improve; sometimes it even pulls resources away from it.

The Dualistic-Development Thesis (4/5): Dualism vs. Lewis; Policy

Dualism vs. Lewis Model

Lewis Model	Dualistic-Development Thesis
Labor moves from the farm to the city Assumes eventual convergence	Sectors coexist but remain divided Assumes persistent divergence

Implications for Development Policy

- ▶ Need targeted support for lagging sectors
- ▶ Infrastructure and education investments in rural and informal areas
- ▶ Inclusive planning beyond GDP growth: the challenge is not just growth, it is **inclusive growth**

The Dualistic-Development Thesis (5/5): Critiques and Case

Critiques of the Thesis

- ▶ Too pessimistic?
- ▶ Ignores potential linkages (supply chains, mobile finance such as M-PESA, skills training)
- ▶ Understates private sector adaptation

Case – India

- ▶ High-tech cities like Bangalore and Hyderabad
- ▶ But also millions working in agriculture with low productivity
- ▶ Dualism even within cities: IT and finance professionals next to informal street vendors and domestic workers

Economic growth does not automatically mean broad progress: dualism can persist for decades, raise inequality, and undermine national development goals.

Dependence Theories: Conclusions and Implications

Shared Perspective of Dependence Theorists

- ▶ Reject the narrow focus on **GDP growth** and **neoclassical models**
- ▶ Criticize:
 - ▶ **Lewis-type models** (e.g. two-sector growth)
 - ▶ **Chenery's structural patterns** as overly rigid or misleading
- ▶ Emphasize the need for:
 - ▶ **Political, institutional, and economic reforms**
 - ▶ Addressing **global power imbalances**

Policy Implications

- ▶ In extreme cases, support:
 - ▶ **Nationalization of assets**
 - ▶ **Public ownership** to fight poverty and inequality
- ▶ Most scholars support:
 - ▶ Reforms combined with **balanced public and private sector roles**
 - ▶ Accelerated growth tied to **social development goals**

Criticism: dependence theory offers little practical guidance, and countries that pursued autarky or full state control fared poorly: China, India, and South Korea grew by opening to

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The Neoclassical Counterrevolution: Free Markets Strike Back

Historical Context

- ▶ 1950s–1970s: development economics dominated by state planning, import substitution, and heavy public investment
- ▶ Historical trigger: economic stagnation, inflation, and debt crises in the 1970s; rise of Reagan/Thatcher
- ▶ Inspired by Hayek, Friedman, Chicago School: the problem is bad policies and too much government

Core Beliefs and Tools

- ▶ Markets are perfect; governments distort
- ▶ Tools: privatization, deregulation, trade liberalization
- ▶ Criticism: ignores inequality, assumes perfect markets

Its opponents' label, “market fundamentalism”, signals the core claim: let the market do its job.

Neoclassical Counterrevolution (1/3): The Free-Market Approach

- ▶ Markets solve problems; government is the problem: prices signal scarcity, profits encourage innovation, competition ensures efficiency
- ▶ Thinkers: Hayek (state planning threatens freedom), Friedman (liberty, price stability, limited government), Chicago School
- ▶ Policies: privatize, deregulate, liberalize — central to the Washington Consensus and structural adjustment programs
- ▶ Criticism: ignores monopolies, externalities, information asymmetry, inequality, and market failures

“Government failure is worse than market failure.”

Neoclassical Counterrevolution (2/3): Public Choice Theory

- ▶ Bureaucrats and politicians are self-interested: economic agents who maximize their own utility (re-election, budgets, the “revolving door”)
- ▶ Government is prone to corruption and **rent-seeking**: special interests lobby for tax exemptions, loopholes, and uncompetitive contracts
- ▶ Shrink government; build in checks and balances: spending limits, independent audits, rules over discretion
- ▶ Thinkers: Buchanan (Nobel 1986), Tullock — *The Calculus of Consent*

“Politics without romance”: do not assume politicians are angels. But the theory underestimates effective states — South Korea and Taiwan fostered development without falling into corruption.

Neoclassical Counterrevolution (3/3): The Market-Friendly Approach

- ▶ Markets work best, but need help: clear rules, functioning courts, basic infrastructure, trust in contracts
- ▶ Government builds institutions: law, property rights, infrastructure, financial systems that serve small firms
- ▶ Key supporter: World Bank's 1991 *World Development Report*, which acknowledged that earlier reforms had focused too much on liberalization
- ▶ Balance between state and market: markets drive growth, governments enable inclusion and stability

A **complementary** role for the state: not big government or no government, but smart government that fixes what the market cannot do on its own.

The Washington Consensus

- ▶ 10 reform principles proposed by John Williamson (1989), summarizing IMF, World Bank, and U.S. Treasury advice
- ▶ Goals: stabilize, liberalize, open markets
- ▶ Used by IMF/World Bank as loan conditions (“conditionality”)
- ▶ Criticism: one-size-fits-all, worsened inequality in some cases

The ten principles

1. Fiscal discipline
2. Reorder public spending (health, education)
3. Tax reform
4. Market-determined interest rates
5. Competitive exchange rates
6. Trade liberalization
7. Liberalize foreign direct investment
8. Privatize state-owned enterprises
9. Deregulate markets
10. Secure property rights

The Washington Consensus: Real-World Impact

- ✓ Chile: market reforms under Pinochet, introduced gradually with competent technocrats → export growth
- ✓ Uganda, Ghana: macro stability restored
- × Zimbabwe: reforms + political crisis = collapse and hyperinflation
- × Latin America: inequality, social unrest

Why the same reforms succeed in one country and fail in another: outcomes depend on institutional capacity, political stability, social protection during reform, and whether reforms are gradual and participatory or sudden and top-down. Economic policies are not “plug and play.”

Criticisms of the Neoclassical Approach

- ▶ Assumes ideal market conditions: perfect competition and information, no externalities — but markets in developing countries are often oligopolistic, information-poor, and constrained by weak legal systems
- ▶ Overemphasizes growth, ignores poverty and distribution: growing the pie does not mean everyone gets a slice; growth $\neq$ development
- ▶ Standardized solutions overlook local context: a policy that works in Chile might fail in Nigeria
- ▶ IMF/World Bank conditions limit sovereignty: reforms not democratically chosen, with strict deadlines, triggering anti-globalization protests

Reforms based on good theory can fail in practice when the assumed conditions do not exist. Successful policy needs to be realistic, adapted, and inclusive.

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Contributions and Limitations of the Four Schools

▶ **Capital Fundamentalism**

- ▶ Points up the importance of investment and of efficiency of capital allocation
- ▶ Investment may be necessary, but it is not sufficient. Broader context matters

▶ **Structural/Empirical Patterns of Development**

- ▶ Careful empirical evidence can remove theories from contention
- ▶ But still need theory to interpret data, avoid cart-before-horse policies

▶ **Dependency**

- ▶ Existing international relations/ trade/ investment can place constraints on the pattern of development
- ▶ But a growing number of counter-examples of stronger versions of dependency theory; good performance of “globally” integrated countries

▶ **Market Fundamentalism**

- ▶ Governments fail (e.g., in SOEs, planning) and we must account for this
- ▶ But markets also fail in developing countries; the East Asia experience shows that the government role can be constructive

Reconciling the Differences (1/3): Key Message

Key Message:

- ▶ No single development theory provides a complete explanation.
- ▶ Each school offers **useful insights**, and a **balanced approach** is needed to understand and guide development policy.

Why? Development is a complex, multi-dimensional process: economic growth, institutional change, political reform, global trade, social equity, cultural shifts. One lens — “more investment,” “more markets,” or “less dependency” — always falls short.

South Korea used state-led planning (industrial targeting, export promotion) *and* market mechanisms; integrated into the global economy *with* protection for key industries; invested heavily in education, not just physical capital. Success came from being adaptive and pragmatic, not from following one model.

Reconciling the Differences (2/3): What Each Theory Contributes

What Each Theory Contributes:

▶ **Linear-Stages Theory**

Highlights the importance of **savings and investment** for long-run growth.

▶ **Structural-Change Models (e.g. Lewis, Chenery)** – Emphasize:

- ▶ Resource transfer from **low- to high-productivity sectors**
- ▶ Empirical evidence of **how economies evolve** over time

▶ **International-Dependence Theories** – Draw attention to:

- ▶ **Global power imbalances**
- ▶ The **vulnerability of developing countries** to decisions made in rich nations and international institutions (e.g., IMF, World Bank)

▶ **Neoclassical Counter-Revolution** – Stresses:

- ▶ **Efficiency of markets** and **price systems**
- ▶ **State failure risks** (e.g., inefficient SOEs, distorted planning)
- ▶ But also needs **adaptation to institutional realities** in developing countries

Reconciling the Differences (3/3): A Balanced Viewpoint

- ▶ **Markets** are important, but **not sufficient**: they allocate resources, reward innovation, and create competition, but underprovide public goods, miss marginalized groups, and can reinforce inequality
- ▶ **Government intervention** is sometimes necessary to:
 - ▶ Correct **market failures** (pollution, monopolies, underinvestment in education and health)
 - ▶ Promote **equity and inclusion** (safety nets, inclusive infrastructure)
- ▶ Effective development requires **context-specific policies** combining:
 - ▶ Market principles (efficiency, incentives)
 - ▶ Public leadership (vision, accountability, coordination)
 - ▶ Institutional awareness (what works here may not work there)

Do not follow ideology — follow evidence and pragmatism.

Chapter summary

- ▶ **Linear-stages theories** (Rostow, Harrod-Domar) treat development as replicating the West's growth path through capital accumulation: $\text{growth} = s/c$ (minus depreciation); saving and investment are necessary but not sufficient.
- ▶ The **Solow model** adds diminishing returns to capital: with a neoclassical production function the economy converges to a steady state $sf(k^*) = (n + \delta)k^*$, and the golden rule picks the saving rate that maximizes consumption.
- ▶ **Structural-change models** (Lewis) explain development as the transfer of surplus labor from a traditional sector to a modern sector whose profits are reinvested; the empirical patterns are stylized facts, not laws.
- ▶ **International-dependence theories** (neocolonial, false-paradigm, dualism) locate underdevelopment in global power imbalances and misapplied foreign models, but offer little practical guidance.
- ▶ The **neoclassical counterrevolution** (free-market, public-choice, market-friendly; the Washington Consensus) stresses market efficiency and government failure, yet markets also fail. A balanced, context-specific approach is needed.

Concepts for Review

- ▶ Autarky
- ▶ Average product
- ▶ Capital-labor ratio
- ▶ Capital-output ratio
- ▶ Center
- ▶ Closed economy
- ▶ *Comprador* groups
- ▶ Dependence
- ▶ Dominance
- ▶ Dualism
- ▶ False-paradigm model
- ▶ Free market
- ▶ Free-market analysis
- ▶ Harrod-Domar growth model
- ▶ Lewis two-sector model
- ▶ Necessary condition
- ▶ Neoclassical counterrevolution
- ▶ Neocolonial dependence model
- ▶ Net savings ratio
- ▶ New political economy approach
- ▶ Open economy
- ▶ Patterns-of-development analysis
- ▶ Periphery
- ▶ Production function
- ▶ Public-choice theory
- ▶ Self-sustaining growth
- ▶ Solow neoclassical growth model
- ▶ Stages-of-growth model of development
- ▶ Structural-change theory
- ▶ Structural transformation